

ELE226 Circuit Theory II Homework 2-3

1 Questions

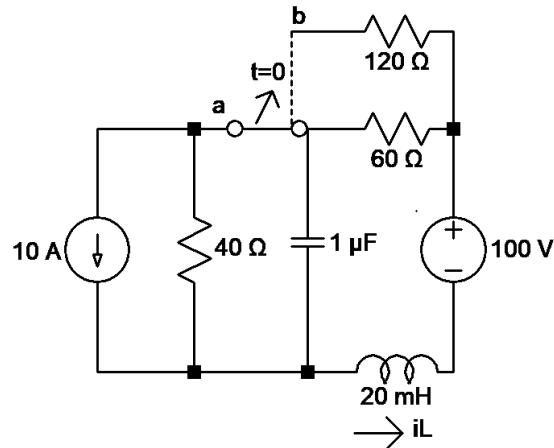
Homework 2

- 2.1. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Assessment Problem 12.2, p. 440*
- 2.2. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.13, p. 460*
- 2.3. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.20, p. 460*
- 2.4. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.22, p. 460*
- 2.5. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.24, p. 461*
- 2.6. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.40, p. 462*
- 2.7. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Assessment Problem 12.5, p. 447*
- 2.8. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.41, p. 462*
- 2.9. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.43, p. 463*
- 2.10. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.43, p. 463*

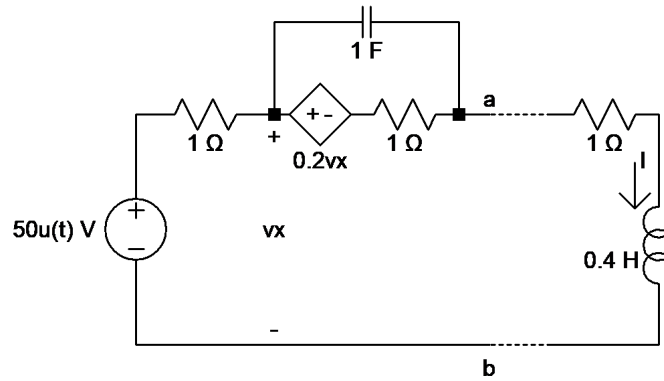
Homework 3

- 3.1. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.27, p. 461*
- 3.2. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.28, p. 461*
- 3.3. *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 12.28, p. 461*

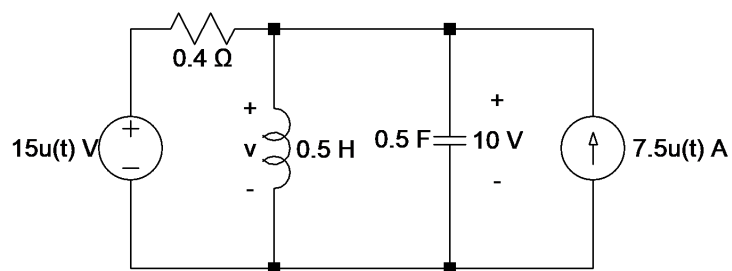
- 3.4.** The switch has been in position a for a long time. At $t = 0$, it moves to position b . Find $i_L(t)$ for $t \geq 0$. Verify i_{Lss} using the phasor approach.



- 3.5.** The capacitor is initially uncharged. Find $I(s)$.



- 3.6.** Find $v(t)$, $t > 0$ using the superposition principle.



- 3.7.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.9, p. 505*
- 3.8.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.11, p. 505*
- 3.9.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.13, p. 506*

- 3.10.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.14, p. 506*
- 3.11.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.17, p. 506*
- 3.12.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.20, p. 507*
- 3.13.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.23, p. 507*
- 3.14.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.28, p. 508*
- 3.15.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.29, p. 508*
- 3.16.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.30, p. 508*
- 3.17.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.32, p. 509*
- 3.18.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.35, p. 509*
- 3.19.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.36, p. 509*
- 3.20.** *J. W. Nilsson and S. Riedel, Electric Circuits, 10th ed., Pearson, 2014, Problem 13.36, p. 509*

2 Answers

Homework 2

2.1. a) $\mathcal{L}\{t^2\} = \frac{2}{s^3} \implies \mathcal{L}\{t^2 e^{-at}\} = \boxed{\frac{2}{(s+a)^3}}.$

b)
$$\begin{aligned} \mathcal{L}\left\{\frac{d}{dt}(e^{-at} \sinh \beta t)\right\} &= s\mathcal{L}\{e^{-at} \sinh \beta t\} - e^{-at} \sinh \beta t|_{t=0} \\ &= s \cdot \frac{\beta}{(s+a)^2 - \beta^2} - 1 \cdot 0 = \boxed{\frac{s\beta}{(s+a)^2 - \beta^2}}. \end{aligned}$$

c) $\mathcal{L}\{t \cos \omega t\} = -\frac{d}{ds} \left(\frac{s}{s^2 + \omega^2} \right) = -\frac{s^2 + \omega^2 - s \cdot 2s}{(s^2 + \omega^2)^2} = \boxed{\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}}.$

2.2. a) Remember: $\mathcal{L}\{g(t-a)u(t-a)\} = e^{-as}F(s)$. Let $g(t) = 20e^{-500t}$. Then

$$\mathcal{L}\{g(t)\} = \frac{20}{s+500} \implies \mathcal{L}\{g(t-a)u(t-a)\} = \boxed{\frac{20e^{-10s}}{s+500}}$$

b) Rearrange the equation:

$$f(t) = u(t+4)(5t+20) + u(t+2)(-10t-20) + u(t-2)(10t-20) + u(t-4)(5t-20)$$

$$\mathcal{L}\{f(t)\} = \boxed{5e^{4s} \cdot \frac{1}{s^2} - 10e^{2s} \cdot \frac{1}{s^2} + 10e^{-2s} \cdot \frac{1}{s^2} + 5e^{-4s} \cdot \frac{1}{s^2}}$$

2.3. a) $\mathcal{L}\{t\} = \frac{1}{s^2} \implies \mathcal{L}\{te^{-at}\} = \boxed{\frac{1}{(s+a)^2}}.$

b) $\mathcal{L}\{f(t)\} = \boxed{\frac{w}{s^2 + \omega^2}}$

c) Use the formula $\sin(a+b) = \sin a \cos b + \cos a \sin b$.

$$\sin(\omega t + \theta) = \sin(\omega t) \cos \theta + \cos(\omega t) \sin \theta \implies \mathcal{L}\{f(t)\} = \boxed{\frac{\omega \cos \theta + s \sin \theta}{s^2 + \omega^2}}$$

d) $\mathcal{L}\{f(t)\} = \boxed{\frac{1}{s^2}}$

e) Use the formula $\cosh(a+b) = \cosh a \cosh b + \sinh a \sinh b$.

$$\cosh(t+\theta) = \cosh t \cosh \theta + \sinh t \sinh \theta \implies \mathcal{L}\{f(t)\} = \boxed{\frac{s \cosh \theta - \sinh \theta}{s^2 - 1}}$$

2.4. a) $\mathcal{L}\left\{\frac{d}{dt}(\sin \omega t)\right\} = s\mathcal{L}\{\sin \omega t\} - \sin \omega t|_{t=0} = s \cdot \frac{\omega}{s^2 + \omega^2} - 0 = \boxed{\frac{s\omega}{s^2 + \omega^2}}.$

b) $\mathcal{L}\left\{\frac{d}{dt}(\cos \omega t)\right\} = s\mathcal{L}\{\cos \omega t\} - \cos \omega t|_{t=0} = s \cdot \frac{s}{s^2 + \omega^2} - 1 = \boxed{\frac{-\omega^2}{s^2 + \omega^2}}.$

c)
$$\begin{aligned} \mathcal{L}\left\{\frac{d^3}{dt^3}(t^2 u(t))\right\} &= s^3 \mathcal{L}\{t^2 u(t)\} - s^2 t^2 u(t)|_{t=0} - s(t^2 u(t))'|_{t=0} - (t^2 u(t))''|_{t=0} \\ &= s^3 \cdot \frac{2}{s^3} - 0 - s(2tu(t) + t^2 \delta(t))|_{t=0} - (2u(t) + 2t\delta(t))|_{t=0} \\ &= \boxed{2}. \end{aligned}$$

$$\text{d) (a)} \rightarrow \mathcal{L}\{\omega \cos \omega t\} = \omega \mathcal{L}\{\cos \omega t\} = \boxed{\frac{s\omega}{s^2 + \omega^2}}.$$

$$\text{(b)} \rightarrow \mathcal{L}\{-\omega \sin \omega t\} = -\omega \mathcal{L}\{\sin \omega t\} = \boxed{\frac{-\omega^2}{s^2 + \omega^2}}.$$

$$\text{(c)} \rightarrow \mathcal{L}\{2\delta(t)\} = 2e^{-0s} = \boxed{2}.$$

$$\begin{aligned} \text{2.5. a) } \mathcal{L}\left\{\frac{d}{dt}(e^{-at} \cos \omega t)\right\} &= s\mathcal{L}\{e^{-at} \cos \omega t\} - e^{-at} \cos \omega t|_{t=0} \\ &= s \cdot \frac{s+a}{(s+a)^2 + \omega^2} - 1 = \boxed{\frac{-as - a^2 - \omega^2}{(s+a)^2 + \omega^2}}. \end{aligned}$$

$$\text{b) } \mathcal{L}\left\{\int_{0^-}^t e^{-ax} \sin \omega x dx\right\} = \frac{\mathcal{L}\{e^{-ax} \sin \omega x\}}{s} = \boxed{\frac{w}{[(s+a)^2 + \omega^2]s}}.$$

$$\text{2.6. a) } F(s) = \frac{K_1}{s+5} + \frac{K_2}{s+8}.$$

$$K_1 = \left. \frac{6(s+10)}{s+8} \right|_{s=-5} = 10, K_2 = \left. \frac{6(s+10)}{s+5} \right|_{s=-8} = -4 \Rightarrow \boxed{f(t) = 10e^{-5t} - 4e^{-8t}}$$

$$\text{b) } F(s) = \frac{K_1}{s} + \frac{K_2}{s+3} + \frac{K_3}{s+7}.$$

$$K_1 = \left. \frac{20s^2 + 141s + 315}{s^2 + 10s + 21} \right|_{s=0} = 15, K_2 = \left. \frac{20s^2 + 141s + 315}{s(s+3)} \right|_{s=-7} = 11$$

$$K_3 = \left. \frac{20s^2 + 141s + 315}{s(s+7)} \right|_{s=-3} = 6 \Rightarrow \boxed{f(t) = 15 - 6e^{-3t} + 11e^{-7t}}$$

$$\text{c) } F(s) = \frac{K_1}{s+2} + \frac{K_2}{s+4} + \frac{K_3}{s+6}.$$

$$K_1 = \left. \frac{15s^2 + 112s + 228}{(s+4)(s+6)} \right|_{s=-2} = 8, K_2 = \left. \frac{15s^2 + 112s + 228}{(s+2)(s+6)} \right|_{s=-4} = -5,$$

$$K_3 = \left. \frac{15s^2 + 112s + 228}{(s+2)(s+4)} \right|_{s=-6} = 12 \Rightarrow \boxed{f(t) = 8e^{-2t} - 5e^{-4t} + 12e^{-6t}}$$

$$\text{d) } F(s) = \frac{K_1}{s} + \frac{K_2}{s+1} + \frac{K_3}{s+2} + \frac{K_4}{s+3}.$$

$$K_1 = \left. \frac{2s^3 + 33s^2 + 93s + 54}{(s+1)(s+2)(s+3)} \right|_{s=0} = 9, K_2 = \left. \frac{2s^3 + 33s^2 + 93s + 54}{s(s+2)(s+3)} \right|_{s=0} = 4$$

$$K_3 = \left. \frac{2s^3 + 33s^2 + 93s + 54}{s(s+1)(s+3)} \right|_{s=0} = -8, K_4 = \left. \frac{2s^3 + 33s^2 + 93s + 54}{s(s+1)(s+2)} \right|_{s=0} = -3$$

$$\Rightarrow \boxed{f(t) = 9 + 4e^{-t} - 8e^{-2t} - 3e^{-3t}}$$

$$\text{2.7. We have } F(s) = \frac{K_1}{s+5} + \frac{K_2}{s+5-j12} + \frac{K_2^*}{s+5+j12}.$$

$$K_1 = \left. \frac{10(s^2 + 119)}{s^2 + 10s + 169} \right|_{s=-5} = 10, K_2 = \left. \frac{10(s^2 + 119)}{(s+5)(s+5+j12)} \right|_{s=-5+j12} = j4.16$$

$$\Rightarrow \boxed{f(t) = 10e^{-5t} + 8.33e^{-5t} \cos(12t + 90^\circ)}$$

$$\mathbf{2.8. \quad a) \quad} F(s) = \frac{K}{s+7-j14} + \frac{K^*}{s+7+j14}.$$

$$K = \frac{280}{s+7-j14} \Big|_{s=-7-j14} = j10 \implies \boxed{f(t) = 20e^{-7t} \cos(14t - 90^\circ)}$$

$$\mathbf{b) \quad} F(s) = \frac{K_1}{s} + \frac{K_2}{s+5-j8} + \frac{K_2^*}{s+5+j8}$$

$$K_1 = \frac{-s^2 + 52s + 445}{s^2 + 10s + 89} \Big|_{s=0} = 5, \quad K_2 = \frac{-s^2 + 52s + 445}{s(s+5+j8)} \Big|_{s=-5+j8} = -3 + j2$$

$$\boxed{f(t) = 5 + 7.21e^{-5t} \cos(8t - 146.31^\circ)}$$

$$\mathbf{c) \quad} F(s) = \frac{K_1}{s+6} + \frac{K_2}{s+2-4j} + \frac{K_2^*}{s+2+4j}.$$

$$K_1 = \frac{14s^2 + 56s + 152}{s^2 + 4s + 20} \Big|_{s=-6} = 10, \quad K_2 = \frac{14s^2 + 56s + 152}{(s+2+j4)(s+6)} \Big|_{s=-2+j4} = 2 + j2$$

$$\implies \boxed{f(t) = 10e^{-6t} + 5.66e^{-2t} \cos(4t + 45^\circ)}$$

$$\mathbf{d) \quad} F(s) = \frac{K_1}{s+5-j3} + \frac{K_1^*}{s+5+j3} + \frac{K_2}{s+4-j2} + \frac{K_2^*}{s+4+j2}$$

$$K_1 = \frac{8(s+1)^2}{(s+5+j3)(s^2+8s+20)} \Big|_{s=-5+j3} = 3.54 - j2.97$$

$$K_2 = \frac{8(s+1)^2}{(s^2+10s+34)(s+4+2j)} \Big|_{s=-4+j2} = -3.54 + j0.69$$

$$\boxed{f(t) = 9.25e^{-5t} \cos(3t - 40.04^\circ) + 7.22e^{-4t} \cos(2t + 168.93^\circ)}$$

$$\mathbf{2.9. \quad a) \quad} F(s) = \frac{K_1}{s^2} + \frac{K_2}{s} + \frac{K_3}{s+8}.$$

$$K_1 = \frac{320}{s+8} \Big|_{s=0} = 40, \quad K_2 = \frac{d}{ds} \left(\frac{320}{s+8} \right) \Big|_{s=0} = -5, \quad K_3 = \frac{320}{s^2} \Big|_{s=-8} = 5$$

$$\boxed{f(t) = 40t - 5 + 5e^{-8t}}$$

$$\mathbf{b) \quad} F(s) = \frac{K_1}{s} + \frac{K_2}{(s+2)^2} + \frac{K_3}{s+2}.$$

$$K_1 = \frac{80(s+3)}{(s+2)^2} \Big|_{s=0} = 60, \quad K_2 = \frac{80(s+3)}{s} \Big|_{s=-2} = -40$$

$$K_3 = \frac{d}{ds} \left[\frac{80(s+3)}{s} \right] \Big|_{s=-2} = -60 \implies \boxed{f(t) = 60 - 40te^{-2t} - 60e^{-2t}}$$

$$\text{c) } F(s) = \frac{K_1}{(s+1)^2} + \frac{K_2}{s+1} + \frac{K_3}{s+3-j4} + \frac{K_4}{s+3+j4}.$$

$$K_1 = \frac{60(s+5)}{s^2+6s+25} \Big|_{s=-1} = 12, \quad K_2 = \frac{d}{ds} \left[\frac{60(s+5)}{s^2+6s+25} \right]_{s=-1} = 0.6$$

$$K_3 = \frac{d}{ds} \left[\frac{60(s+5)}{(s+1)^2(s+3+j4)} \right]_{s=-3+j4} = -0.3 + j1.65$$

$$\Rightarrow \boxed{f(t) = 12te^{-t} + 0.6e^{-t} + 3.35e^{-3t} \cos(4t + 100.305^\circ)}$$

$$\text{d) } F(s) = \frac{K_1}{(s+5)^2} + \frac{K_2}{s+5} + \frac{K_3}{s^2} + \frac{K_4}{s}.$$

$$K_1 = \frac{25(s+4)^2}{s^2} \Big|_{s=-5} = 1, \quad K_2 = \frac{d}{ds} \left[\frac{25(s+4)^2}{s^2} \right]_{s=-5} = -1.6$$

$$K_3 = \frac{25(s+4)^2}{(s+5)^2} \Big|_{s=-5} = 16, \quad K_4 = \frac{d}{ds} \left[\frac{25(s+4)^2}{(s+5)^2} \right]_{s=-5} = 1.6$$

$$\Rightarrow \boxed{f(t) = 1.6 + 16t + te^{-5t} - 1.6e^{-5t}}$$

$$\text{2.10. a) } F(s) = \frac{K_1}{s} + \frac{K_2}{(s+3)^3} + \frac{K_3}{(s+3)^2} + \frac{K_4}{s+3}.$$

$$K_1 = \frac{135}{(s+3)^3} \Big|_{s=0} = 5, \quad K_2 = \frac{135}{s} \Big|_{s=-3} = -45$$

$$K_3 = \frac{d}{ds} \left(\frac{135}{s} \right) \Big|_{s=-3} = -15, \quad 2K_4 = \frac{d^2}{ds^2} \left(\frac{135}{s} \right) \Big|_{s=-3} = -10. \Rightarrow K_4 = -5$$

$$\boxed{f(t) = 5 - 5e^{-3t} - 15te^{-3t} - 22.5t^2e^{-3t}}$$

$$\text{b) } F(s) = \frac{As+B}{(s^2+2s+2)^2} + \frac{Cs+D}{s^2+2s+2}.$$

$$10(s+2)^2 \Big|_{s=-1+j} = -j20 = A(-j+1) + B \Rightarrow A = 20, B = 20$$

$$\frac{d}{ds} [10(s+2)^2] \Big|_{s=-1+j} = j20 + 20 = A + C(s^2+2s+2) + (Cs+D)(2s+2) \Big|_{s=-1+j}$$

$$\Rightarrow C = 0, D = 10.$$

$$\boxed{f(t) = 10(t+1)e^{-t} \sin t}$$

$$\text{c) } F(s) = 25 + \frac{20s+114}{s^2+15s+54} = 25 + \frac{K_1}{s+9} + \frac{K_2}{s+6}$$

$$K_1 = \frac{20s+144}{s+6} \Big|_{s=-9} = 12, \quad K_2 = \frac{20s+144}{s+9} \Big|_{s=-6} = 8$$

$$\Rightarrow \boxed{f(t) = 25\delta(t) + 8e^{-6t} + 12e^{-9t}}$$

$$\text{d) } F(s) = 5s - 15 + \frac{K_1}{s+2} + \frac{K_2}{s+5}$$

$$K_1 = \left. \frac{6s+42}{s+5} \right|_{s=-2} = 10, \quad K_2 = \left. \frac{6s+42}{s+2} \right|_{s=-5} = -4$$

$$\Rightarrow \boxed{f(t) = 5\delta'(t) - 15\delta(t) + 10e^{-2t} - 4e^{-5t}}$$

Homework 3

3.1. a. $\boxed{-I_{dc} + \frac{1}{L} \int_0^t v_o(\tau) d\tau + \frac{v_o}{R} + C \frac{dv_o}{dt} = 0}$

b. $\frac{-I_{dc}}{s} + \frac{1}{L} \cdot \frac{V_o}{s} + \frac{V_o}{R} + CsV_o + \cancel{CV_o(0)} = 0 \Rightarrow \boxed{V_o(s) = \frac{I_{dc}/C}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \text{ V-s}}$

c. $i_o = C \frac{dv_o}{dt} \Rightarrow \boxed{I_o(s) = sCV_o(s) = \frac{sI_{dc}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \text{ A-s}}$

3.2. a. $\boxed{\frac{v_o}{R} + C \frac{dv_o}{dt} + \frac{1}{L} \int_0^t v_o(\tau) d\tau = 0}$

b. We have $\frac{L}{R} \frac{di_o}{dt} + LC \frac{d^2 i_o}{dt^2} + i_o = 0$.

$$\frac{L}{R} sI_o - \frac{L}{R} I_{dc} + LC(s^2 I_o - sI_{dc} - 0) + I_o = 0 \Rightarrow I_o \left(\frac{sL}{R} + LCs^2 + 1 \right) - I_{dc} \left(\frac{L}{R} + sLC \right)$$

$$\Rightarrow \boxed{I_o(s) = \frac{I_{dc} \left(s + \frac{1}{RC} \right)}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \text{ A-s}}$$

3.3. a. Let i_o circulate in the clockwise direction.

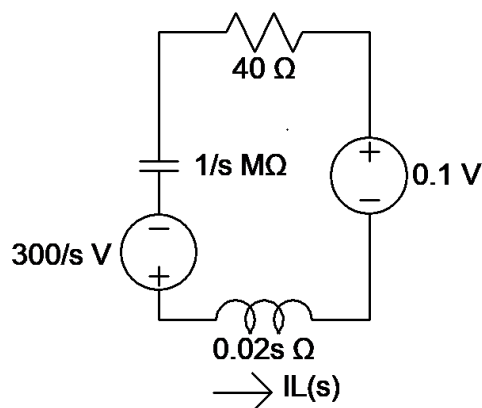
$$Ri_o + L \frac{di_o}{dt} + v_o = 0 \quad \left[i_o = C \frac{dv_o}{dt} \right] \Rightarrow \boxed{\frac{d^2 v_o}{dt^2} + \frac{R}{L} \frac{dv_o}{dt} + \frac{1}{LC} v_o = 0}$$

b. From KVL,

$$s^2 V_o - sV_{dc} - \cancel{v_o'(0)} + \frac{R}{L}(sV_o - V_{dc}) + \frac{1}{LC} V_o = 0 \Rightarrow \boxed{V_o(s) = \frac{V_{dc} \left(s + \frac{R}{L} \right)}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \text{ V-s}}$$

3.4. Initial conditions:

$$i_L(0^-) = 5 \text{ A}, \quad v_C(0^-) = -200 \text{ V}.$$



From KVL,

$$-0.1 - \frac{300}{s} - 40I_L + \frac{10^6}{s}I_L + 20sI_L \cdot 10^{-3} = 0 \implies I_L = \frac{0.1 + \frac{300}{s}}{\frac{10^6}{s} + 40 + 0.02s}$$

$$I_L = \frac{5s + 15000}{s^2 + 2000s + 50 \cdot 10^6} = \frac{K_1}{s + 1000 - j7000} + \frac{K_1^*}{s + 1000 + j7000}$$

$$K_1 = \left. \frac{5s + 15000}{s + 1000 + j7000} \right|_{s=-1000-j7000} \quad z = 2.5 - j0.714$$

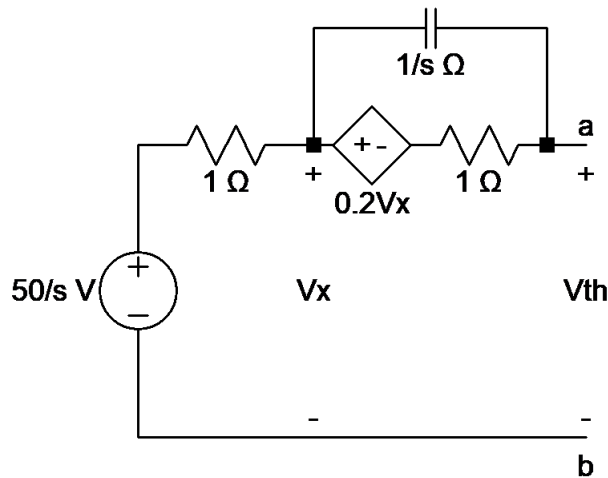
$$\implies \boxed{i_L(t) = [5.2e^{-1000t} \cos(7000t - 15.945^\circ)]u(t)}$$

The steady part is 0 because the capacitor is open at infinity ($\omega = 0 \implies \frac{1}{j\omega C} \rightarrow \infty$), leading to $i_{Lss} = 0$.

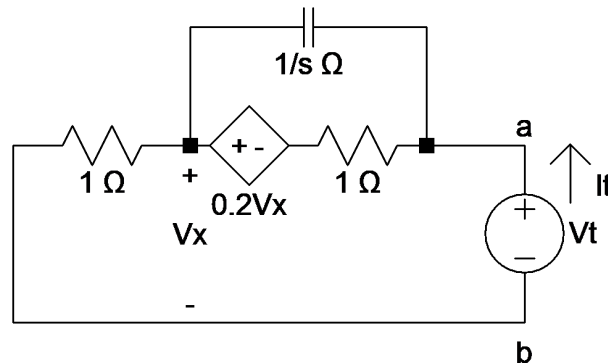
3.5. Calculate V_{Th} .

$$\frac{V_x - \frac{50}{s}}{1} + \frac{V_x - V_{Th}}{1/s} + \frac{V_x - 0.2V_x - V_{Th}}{1} = 0 \quad \text{and} \quad V_x = \frac{50}{s}$$

$$\implies V_{th} = \frac{50s + 40}{s + s^2}$$



Zero the 50 V source and apply a test voltage.

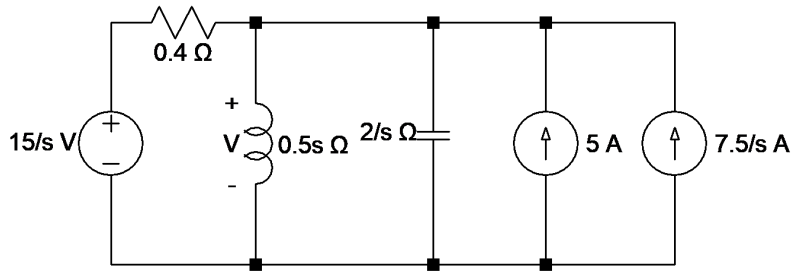


$$\frac{V_x}{1} + \frac{V_x - 0.2V_x - V_t}{1} + \frac{V_x - V_t}{1/s} = 0 \quad \text{and} \quad V_x = i_t$$

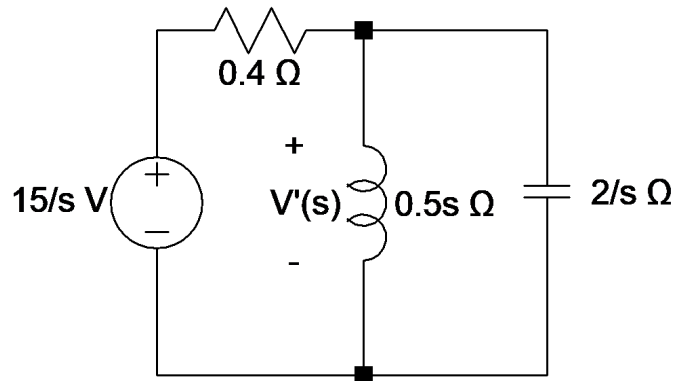
$$I_t(2 - 0.2 + s) = V_t(s + 1) \implies \frac{V_t}{I_t} = R_{Th} = \frac{1.8 + s}{s + 1}$$

$$I(s) = \frac{\frac{50s + 40}{s(s + 1)}}{1 + 0.4s + \frac{s + 1.8}{s + 1}} = \frac{125s + 100}{s(s^2 + 6s + 7)}$$

3.6. The s -domain equivalent circuit:



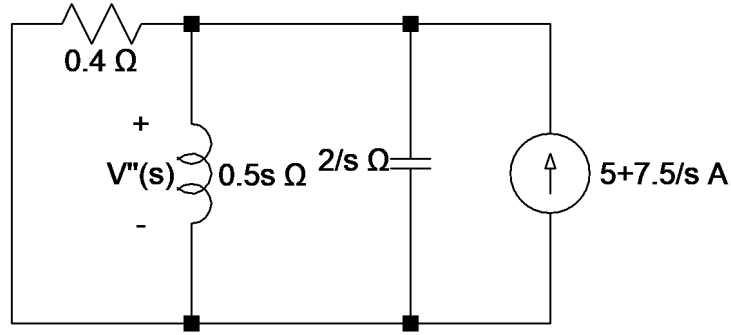
Zero the current source.



$$V'(s) = \frac{15}{s} \cdot \frac{0.5s \parallel \frac{2}{s}}{\left(0.5s \parallel \frac{2}{s}\right) + 0.4} = \frac{75}{(s + 1)(s + 4)} = \frac{K_1}{s + 1} + \frac{K_2}{s + 4}$$

$$K_1 = \left. \frac{75}{s + 4} \right|_{s=-1} = 25, \quad K_2 = \left. \frac{75}{s + 1} \right|_{s=-4} = -25 \implies V'(s) = \frac{25}{s + 1} - \frac{25}{s + 4}$$

Zero the voltage source.

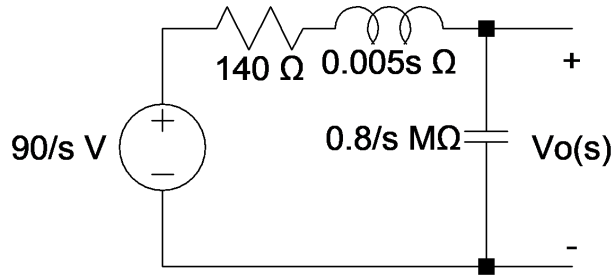


$$V''(s) = \left(\frac{7.5}{s} \parallel s \right) \parallel \left(\frac{2}{s} \parallel 0.5s \parallel 0.4 \right) = \frac{2s}{s^2 + 5s + 4} = \frac{K_1}{s+4} + \frac{K_2}{s+1}$$

$$K_1 = \frac{15 + 10s}{s+1} \Big|_{s=-4} = \frac{25}{3}, \quad K_2 = \frac{15s + 10}{s+4} \Big|_{s=-1} = \frac{5}{3} \Rightarrow V''(s) = \frac{25}{3(s+4)} + \frac{5}{3(s+1)}$$

$$V(s) = V'(s) + V''(s) = -\frac{50}{3} \frac{1}{s+4} + \frac{80}{3} \frac{1}{s+1} \Rightarrow \boxed{v(t) = \left(-\frac{50}{3} e^{-4t} + \frac{80}{3} e^{-t} \right) u(t) \text{ V}}$$

3.7. Find the s -domain circuit.



$$V_o(s) = \frac{90}{s} \cdot \frac{800000/s}{\frac{s}{800000} + 0.005s + 140} = \frac{90}{s} \cdot \frac{16 \cdot 10^7}{s^2 + 28000s + 16 \cdot 10^7}$$

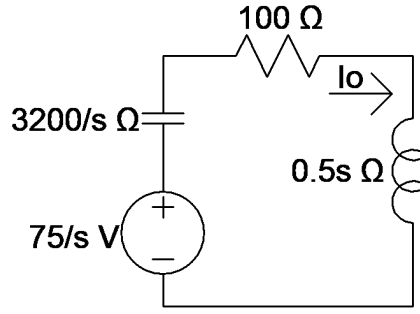
$$= \frac{144 \cdot 10^8}{s(s+20000)(s+8000)} = \frac{K_1}{s} + \frac{K_2}{s+20000} + \frac{K_3}{s+8000}$$

$$K_1 = \frac{144 \cdot 10^9}{(s+2000)(s+8000)} \Big|_{s=0} = 90, \quad K_2 = \frac{144 \cdot 10^9}{s(s+8000)} \Big|_{s=-20000} = 60$$

$$K_3 = \frac{144 \cdot 10^9}{s(s+20000)} \Big|_{s=-8000} = -150$$

$$\Rightarrow \boxed{v_o(t) = [90 - 150e^{-8000t} + 60e^{-20000t}]u(t) \text{ V}}$$

3.8. a) The circuit:



$$\text{b) } -\frac{75}{s} + I_o \frac{3200}{s} + 100I_o + 0.5sI_o = 0 \Rightarrow I_o(s) = \frac{150}{(s+40)(s+160)} \text{ A-s}$$

$$\text{c) } I_o(s) = \frac{K_1}{s+40} + \frac{K_2}{s+160}$$

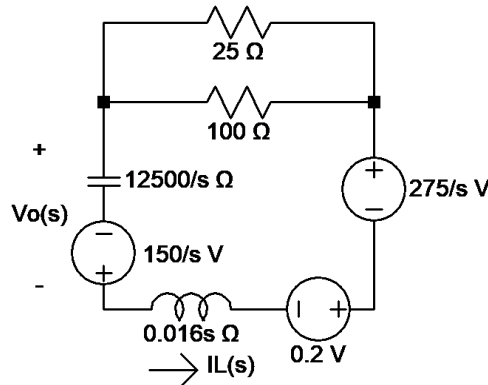
$$K_1 = \left. \frac{150}{s+160} \right|_{s=40} = \frac{5}{4}, \quad K_2 = \left. \frac{150}{s+40} \right|_{s=-160} = -\frac{5}{4}$$

$$\Rightarrow i_o(t) = \frac{5}{4} (e^{-40t} - e^{-160t}) u(t) \text{ A}$$

3.9. Find the current through the inductor.

$$-1.25 + \frac{v_C}{80} + \frac{v_C}{200} + \frac{v_C - 275}{100} = 0 \Rightarrow v_C = 150 \text{ V}, \quad i_L = \frac{275 - 150}{100} = 1.25 \text{ A}$$

a) The circuit:



b) From KCL,

$$0.016sI_L - 0.02 - \frac{2.75}{s} + 20I_L + \frac{12500}{s}I_L + \frac{150}{s} = 0 \Rightarrow I_L = \frac{7812.5 + 1.25s}{s^2 + 1250s + 781250}$$

$$V_o = \frac{12500}{s}I + \frac{150}{s} = \frac{150s^2 + 203125s + 214843750}{s(s^2 + 1250s + 781250)}$$

c) From (b)

$$V_o = \frac{K_1}{s} + \frac{K_2}{s+625-j625} + \frac{K_2^*}{s+625+j625}$$

$$K_1 = \left. \frac{150s^2 + 203125s + 214843750}{s^2 + 1250s + 781250} \right|_{s=0} = 275$$

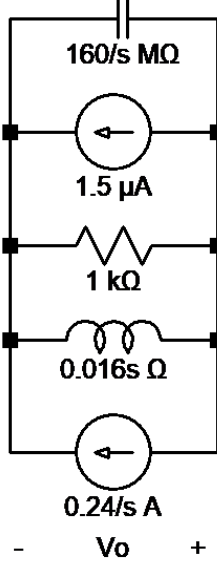
$$K_2 = \frac{150s^2 + 203125 + 214843750}{s(s + 625 + j625)} \Big|_{s=-625+j625} = 80.04/141.34^\circ$$

$$\Rightarrow \boxed{v_o(t) = [2755 + 160.08e^{-625t} \cos(625t + 141.34^\circ)]u(t) \text{ V}}$$

3.10. Initial conditions:

$$v_c(0^-) = v_c(0) = 360 \cdot \frac{4 \text{ k} \parallel 1 \text{ k}}{4 \text{ k} \parallel 1 \text{ k} + 400} = 240 \text{ V}, \quad i_L(0^-) = i_L(0) = \frac{240}{1 \text{ k}} = 0.24 \text{ A}$$

a) The circuit:



b) $Z_{eq} = \frac{16 \cdot 10^7}{s} \parallel 1000 \parallel 0.016s, \quad I_{eq} = 1.5 \cdot 10^{-6} + \frac{0.24}{s}$

$$V_o = -I_{eq}Z_{eq} = -\frac{1.5 \cdot 10^{-6} + \frac{0.24}{s}}{\frac{1}{\frac{16 \cdot 10^7}{s}} + \frac{1}{1000} + \frac{1}{0.016s}} = \boxed{-\frac{240s + 38.4 \cdot 10^6}{s^2 + 160000s + 10^{10}} \text{ V-s}}$$

c) $I_L = \frac{V_o}{0.016s} + \frac{0.24}{s} = \boxed{\frac{0.24s + 23400}{s^2 + 160000s + 10^{10}} \text{ A-s}}$

d) From (b), we have

$$V_o = -\frac{240s + 38.4 \cdot 10^6}{s^2 + 160000s + 10^{10}} = \frac{K_1}{s + 80000 - j60000} + \frac{K_1^*}{s + 80000 + j60000}$$

$$K_1 = \frac{-240s - 38400000}{s + 80000 + j60000} \Big|_{s=-80000+j60000} = 200/126.87^\circ$$

$$\Rightarrow \boxed{v_o(t) = [400e^{-80000t} \cos(60000t + 126.87^\circ)]u(t) \text{ A}}$$

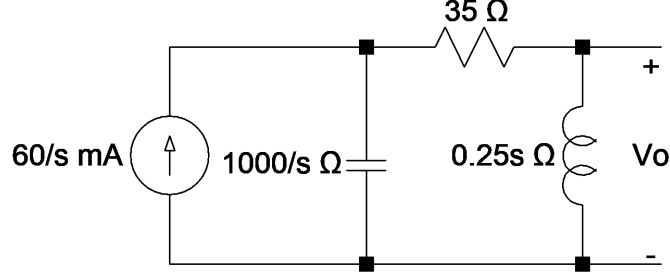
e) From (b), we have

$$I_o = \frac{0.24s + 23400}{s^2 + 160000s + 10^{10}} = \frac{K_1}{s + 80000 - j60000} + \frac{K_1^*}{s + 80000 + j60000}$$

$$K_1 = \frac{0.24s + 23400}{s + 80000 + j60000} \bigg|_{s=-80000+j60000} = 0.125 \angle -16.26^\circ$$

$$\Rightarrow i_L(t) = [0.5e^{-80000t} \cos(60000t - 16.26^\circ)]u(t) \text{ A}$$

3.11. The circuit in the s -domain:



$$\text{a) } V_o = \frac{60}{s^2} \cdot \frac{0.25s}{\frac{1600}{s} + 35 + 0.25s} = \frac{60}{s^2 + 140s + 4000}$$

$$\text{b) } \lim_{t \rightarrow 0} v_o(t) = \lim_{s \rightarrow \infty} sV_o(s) = \frac{s}{s^2 + 140s + 4000} = 0$$

$$\lim_{t \rightarrow \infty} v_o(t) = \lim_{s \rightarrow 0} sV_o(s) = \frac{s}{s^2 + 140s + 4000} = 0.$$

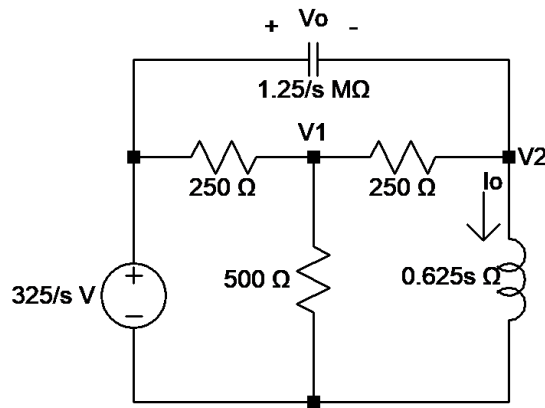
$$v_o(0) = v_o(\infty) = 0$$

c) From (a), we have

$$V_o = \frac{60}{s^2 + 140s + 4000} = \frac{K_1}{s + 100} + \frac{K_2}{s + 40} \quad [K_1 = -1, K_2 = 1 \text{ directly}]$$

$$v_o(t) = (e^{-40t} - e^{-100t})u(t) \text{ V}$$

3.12. The circuit in the s -domain:



a) From KCL equations,

$$\frac{V_1 - \frac{325}{s}}{250} + \frac{V_1}{500} + \frac{V_1 - V_2}{250} = 0, \quad \frac{V_2}{0.625s} + \frac{V_2 - V_1}{250} + \frac{V_2 - \frac{325}{s}}{1250000s} = 0 \Rightarrow 5V_1 - 2V_2 = \frac{650}{s}$$

$$-5000sV_1 + (s^2 + 5000s + 2000000)V_2 = 325s \Rightarrow V_2 = \frac{325}{s + 1000}$$

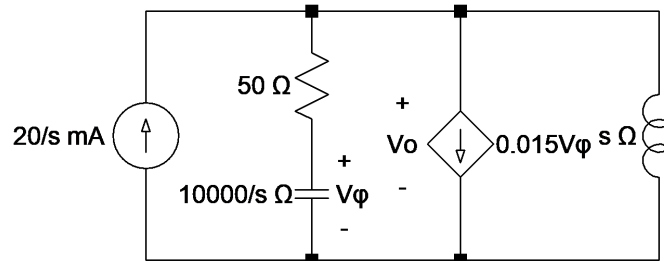
$$\begin{aligned} V_o &= \frac{325}{s} - \frac{325}{s+1000} \\ I_o &= \frac{0.52}{s} - \frac{0.52}{s+1000} \end{aligned}$$

b) Using the results in (a),

$$\begin{aligned} v_o(t) &= (325 - 325e^{-1000t})u(t) \text{ V} \\ i_o(t) &= (520 - 520e^{-1000t})u(t) \text{ mA} \end{aligned}$$

c) The poles of functions of s lie on the y -axis and on the left of the axis, leading to constant and decaying exponential functions of t , which is feasible.

3.13. The circuit in the s -domain:



From KCL,

$$\frac{0.02}{s} + \frac{V_o}{50 + 10^4/s} + 0.015V_\phi + \frac{V_o}{s} = 0.$$

We have $V_\phi = \frac{\frac{10^4}{s}}{50 + \frac{10^4}{s}} V_o = \frac{10^4 \cdot V_o}{50s + 10^4}$

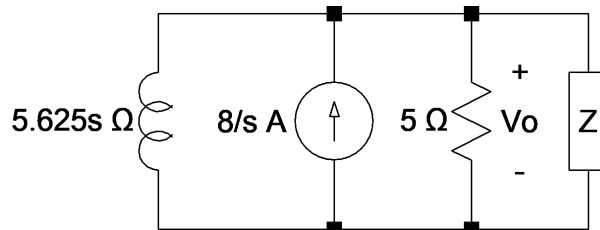
Replacing V_ϕ in the KCL equation and solving for V_o , we get

$$V_o = \frac{s + 200}{s^2 + 200s + 10^4} = \frac{K_1}{(s + 100)^2} + \frac{K_2}{s + 100}$$

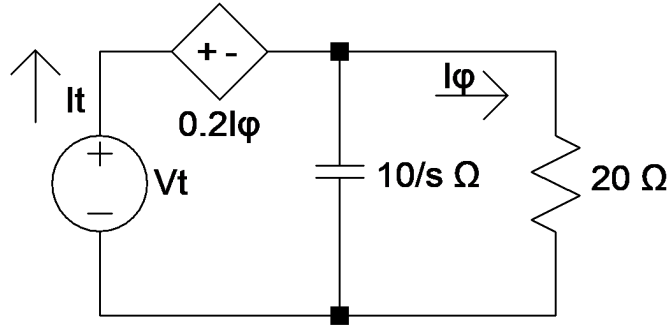
$$K_1 = (s + 200)|_{s=-100} = 100, \quad K_2 = 1$$

$$\Rightarrow v_o(t) = (100te^{-100t} + e^{-100t})u(t) \text{ V}$$

3.14. The s -domain equivalent circuit:



a) Apply a test voltage to find the impedance to the right.



$$V_t = 25I_\phi + \frac{20 \cdot \frac{10}{s}}{20 + \frac{10}{s}} I_t, \quad I_\phi = \frac{1 + \frac{10}{s}}{20 + \frac{10}{s}} \Rightarrow Z = \frac{V_t}{I_t} = \frac{250 + 200}{20s + 10} = \frac{45}{2s + 1}$$

$$\frac{V_o}{s} + \frac{V_o(2s + 1)}{45} + \frac{V_o}{5.625s} = \frac{8}{s} \Rightarrow V_o = \frac{180}{s^2 + 5s + 4} \text{ V-s}$$

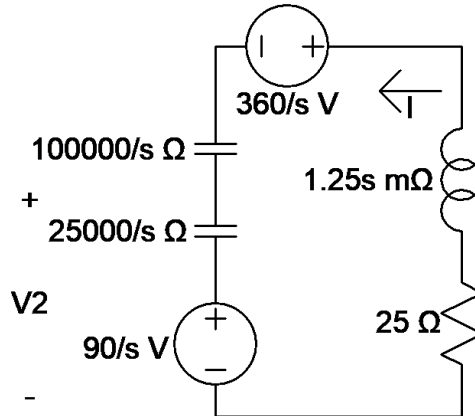
$$V_1 = \left. \frac{180}{s + 1} \right|_{s=-4} = -60, \quad V_2 = \left. \frac{180}{s + 4} \right|_{s=-1} = 60 \Rightarrow \boxed{V_o = \frac{60}{s + 1} - \frac{60}{s + 4} \text{ V-s}}$$

b) $\boxed{v_o(t) = (60e^{-t} - 60e^{-4t})u(t) \text{ V}}$

3.15. Initially, we have

$$v_2(0^-) = 450 \cdot \frac{24 \mu \parallel 16 \mu}{24 \mu \parallel 16 \mu + 10 \mu} = 90 \text{ V}$$

a) The circuit:



b) Let I be the current in the counterclockwise direction.

$$25I + 1.25 \cdot 10^{-3}sI + \frac{450}{s} = 0 \Rightarrow I = \frac{-450s}{25 + 1.25 \cdot 10^{-3}s + \frac{125000}{s}}$$

$$\boxed{V_1 = -25I = \frac{9000000}{(s + 10000)^2}}$$

$$K_1 = 90s + 1800000|_{s=-10000} = 900000, \quad K_2 = 90$$

$$\Rightarrow \boxed{v_1(t) = (9000000te^{-10000t})u(t) \text{ V}}$$

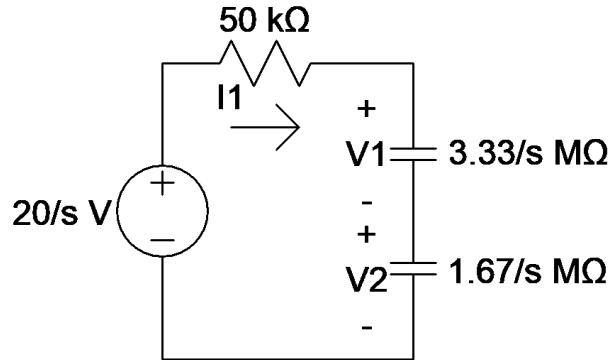
$$\text{c) } v_2 = I \cdot (25000/s) + 90/s = \frac{90s + 1800000}{s^2 + 20000s + 10^8} = \frac{K_1}{(s + 10000)^2} + \frac{K_2}{s + 10000}$$

$$K_1 = 90s + 18 \cdot 10^5|_{s=-10000} = 9 \cdot 10^5, \quad K_2 = 90$$

$$V_2 = \frac{900000}{(s + 10000)^2} + \frac{90}{s + 10000}$$

$$\Rightarrow v_2(t) = [900000te^{-10000t} + 90e^{-10000t}]u(t) \text{ V}$$

3.16. a) The circuit:



b) From KVL,

$$-\frac{20}{s} + 50000I_1 + \frac{5 \cdot 10^6 I_1}{s} = 0 \Rightarrow I_1 = \frac{20}{5000000 + 50000s} = \frac{0.4 \cdot 10^{-3}}{s + 100}$$

$$V_1 = \frac{10}{3} \cdot \frac{10^6}{s} \cdot I = \frac{4000/3}{s(s + 100)}, \quad V_2 = \frac{10^7}{6s} \cdot I = \frac{2000/3}{s(s + 100)}$$

$$\text{c) } I = (0.4 \cdot 10^{-3} e^{-100t})u(t) \text{ A}$$

$$V_1 = \frac{K_1}{s} + \frac{K_2}{s + 100} \Rightarrow K_1 = \frac{4000/3}{s + 100} \Big|_{s=0} = \frac{40}{3}, \quad K_2 = \frac{4000/3}{s} \Big|_{s=-100} = -\frac{40}{3}$$

$$V_1 = \frac{40}{3}(1 - e^{-100t})u(t) \text{ V}, \quad V_2 = \frac{V_1}{2} = \frac{20}{3}(1 - e^{-100t})u(t) \text{ V}$$

d) The poles of functions of s lie on the y -axis and on the left of the axis, leading to constant and decaying exponential functions of t , which is physically meaningful.

3.17. a) The Laplace transform of $V_m \sin(\omega t + \phi)$:

$$\mathcal{L}\{V_m \sin(\omega t + \phi)\} = V_m \mathcal{L}\{\sin \omega t \cos \phi + \cos \omega t \sin \phi\} = \frac{\omega \cos \phi + s \sin \phi}{s^2 + \omega^2}$$

$$I(s) = \frac{V_m \frac{\omega \cos \phi + s \sin \phi}{s^2 + \omega^2}}{sL + R} = \frac{\frac{V_m}{L}(\omega \cos \phi + s \sin \phi)}{(s + \frac{R}{L})(s^2 + \omega^2)} = \frac{K_1}{s + \frac{R}{L}} + \frac{K_2}{s - j\omega} + \frac{K_2^*}{s + j\omega}$$

$$K_1 = \left. \frac{\frac{V_m}{L}(\omega \cos \phi + s \sin \phi)}{s^2 + \omega^2} \right|_{s=-\frac{R}{L}} = \frac{V_m(L\omega \cos \phi - R \sin \phi)}{R^2 + \omega^2 L^2}$$

$$K_2 = \left. \frac{\frac{V_m}{L}(\omega \cos \phi + s \sin \phi)}{(s + \frac{R}{L})(s + j\omega)} \right|_{s=j\omega} = \frac{\frac{V_m}{L}(\omega \cos \phi + j\omega \sin \phi)}{(j\omega + \frac{R}{L})2j\omega} = \frac{V_m(\cos \phi + j \sin \phi)}{(j\omega L + R)2j}$$

$$K_2 = \frac{V_m e^{j\phi}}{2 \angle 90^\circ \sqrt{\omega^2 L^2 + R^2} \angle \tan^{-1}(\omega L/R)} = \frac{V_m}{2\sqrt{\omega^2 L^2 + R^2}} \angle \phi - 90^\circ - \theta(\omega)$$

$$\Rightarrow i(t) = \underbrace{\frac{V_m(L\omega \cos \phi - R \sin \phi)}{\omega^2 L^2 + R^2} e^{-\frac{R}{L}t}}_{\text{Transient State}} + \underbrace{\frac{V_m \sin[\omega t + \phi - \theta(\omega)]}{\sqrt{\omega^2 L^2 + R^2}}}_{\text{Steady State}} \text{ A}$$

b) $\boxed{\frac{V_m(L\omega \cos \phi - R \sin \phi)}{\omega^2 L^2 + R^2} e^{-\frac{R}{L}t}}$

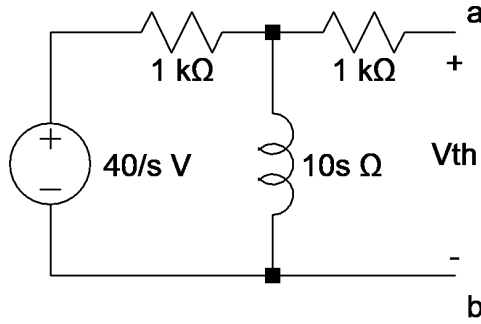
c) $\boxed{\frac{V_m \sin[\omega t + \phi - \theta(\omega)]}{\sqrt{\omega^2 L^2 + R^2}}}$

d) $\mathbf{V}_g = \mathbf{I}(j\omega L + R) \Rightarrow \mathbf{I} = \frac{V_m \angle \phi - 90^\circ}{\sqrt{\omega^2 L^2 + R^2} \angle \theta(\omega)}$

$$\Rightarrow \Re(\mathbf{I}) = \frac{V_m}{\sqrt{\omega^2 L^2 + R^2}} \sin[\omega t + \phi - \theta(\omega)]$$

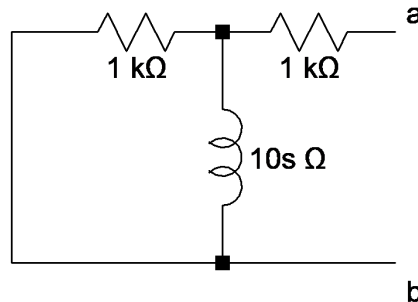
e) $L\omega \cos \phi = R \sin \phi \Rightarrow \phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$

3.18. Calculate V_{th} .



$$V_{th} = \frac{40}{s} \cdot \frac{10s}{10s + 1000} = \frac{40}{s + 100} \text{ V-s}$$

Calculate Z_{th} .



$$Z_{th} = 1000 + \frac{1}{\frac{1}{1000} + \frac{1}{10s}} = \frac{2000s + 100000}{s + 100}$$

$$I = \frac{\frac{40}{s+100}}{\frac{500000}{s} + \frac{2000s+1000}{s+100}} = \frac{40s}{2000s^2 + 600000s + 5000000} = \frac{0.02s}{s^2 + 30s + 25000}$$

$$I = \frac{K_1}{s + 150 - j50} + \frac{K_1^*}{s + 150 + j50}$$

$$\Rightarrow K_1 = \left. \frac{0.02s}{s + 150 + j50} \right|_{s=-150+j50} = 31.62 \cdot 10^{-3} \angle 71.57^\circ$$

$$\Rightarrow \boxed{i(t) = [63.24e^{-150t} \cos(50t + 71.57^\circ)]u(t) \text{ mA}}$$

3.19. a) Let I_1 and I_2 circulate clockwise in the left and right meshes, respectively.

$$(100 + 0.1875s)I_1 + 0.125sI_2 - \frac{150}{s} = 0 \quad \text{and} \quad 0.125sI_1 + (200 + 0.25s)I_2 = 0$$

Solve the latter equation for I_2 and substitute into the former equation, then

$$\Rightarrow \boxed{I_1(s) = \frac{1200(s + 800)}{s(s^2 + 2000s + 640000)} \text{ A-s}}$$

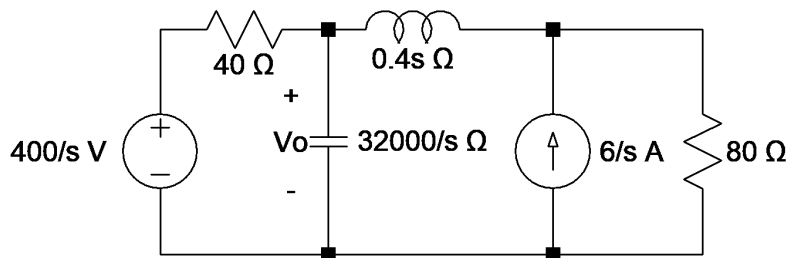
b) $\boxed{i_1(0) = \lim_{s \rightarrow \infty} sI_1 = 0, \quad \lim_{t \rightarrow \infty} i_1(t) = \lim_{s \rightarrow 0} sI_1 = 1.5 \text{ A}}$

c) $I_1 = \frac{K_1}{s} + \frac{K_2}{s + 400} + \frac{K_3}{s + 1600}$

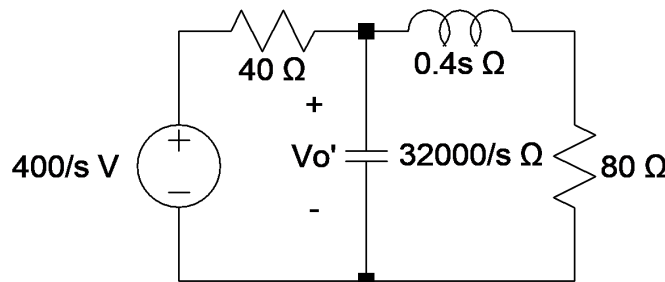
$$K_1 = \left. \frac{1200(s + 800)}{(s^2 + 1000s + 640000)} \right|_{s=0} = 1.5, \quad K_2 = \left. \frac{1200(s + 800)}{s(s + 1600)} \right|_{s=-400} = -1,$$

$$K_3 = \left. \frac{1200(s + 800)}{s(s + 400)} \right|_{s=-1600} = -0.5 \Rightarrow \boxed{i_1(t) = (1.5 - e^{-400t} - 0.5e^{-1600t})u(t) \text{ A}}$$

3.20. a) The s -domain equivalent circuit:

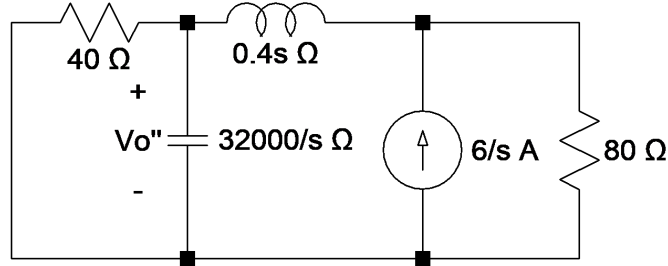


Zero the current source to find the individual response of the voltage source.



$$V'_o = \frac{400}{s} \cdot \frac{\frac{(0.4s + 80) \cdot \frac{32000}{s}}{0.4s + 80 + \frac{32000}{s}}}{40 + \frac{(0.4s + 80) \cdot \frac{32000}{s}}{0.4s + 80 + \frac{32000}{s}}} = \frac{320000(s + 200)}{s(s^2 + 1000s + 240000)}$$

Zero the voltage source.



$$I'' = \frac{80}{\left(\frac{32000}{s} \parallel 40\right) + 0.4s + 80} \cdot \frac{6}{s} = \frac{480(s + 800)}{s(0.4s^2 + 400s + 96000)}$$

$$V_o'' = \frac{1200(s + 800)}{s(s^2 + 1000s + 240000)} \cdot \left(\frac{32000}{s} \parallel 40\right) = \frac{38400000}{s(s^2 + 1000s + 240000)}$$

$$\Rightarrow \boxed{V_o = V'_o + V_o'' = \frac{320000(s + 320)}{s(s^2 + 1000s + 240000)}}$$

b) $V_o = \frac{K_1}{s} + \frac{K_2}{s + 400} + \frac{K_3}{s + 600}.$

$$K_1 = \frac{320000(s + 320)}{s^2 + 1000s + 240000} \Big|_{s=0} = \frac{1280}{3}, \quad K_2 = \frac{320000(s + 320)}{s(s + 600)} \Big|_{s=-400} = 320$$

$$K_3 = \frac{320000(s + 320)}{s(s + 400)} \Big|_{s=600} = -\frac{2240}{3}$$

$$\Rightarrow \boxed{V_o(t) = \left(\frac{1280}{3} + 320e^{-400t} - \frac{2240}{3}e^{-600t}\right)u(t) \text{ V}}$$